# CP2 Histology: top 50 questions

## Topic 9A · Peripheral blood (H11A)

### Centrifuged blood, hematocrit, plasma vs serum

**1.** *2026 TA review*
A centrifuged blood specimen demonstrates a hematocrit of 45% and a negligible buffy coat. Approximately what percentage of the blood volume is plasma?

**A.** 25%
**B.** 35%
**C.** 45%
**D.** 55%
**E.** 75%

**Answer: D. 55%**

Spun blood has three layers: **RBCs** (red blood cells) at the bottom = the hematocrit, a thin **buffy coat** (white cells + platelets, under 1%), and **plasma** on top. Plasma = 100% − 45% hematocrit − about 0% buffy coat = **55%**.
**A (25%):** would need a hematocrit of 75%, far above any normal value.
**B (35%):** would need a hematocrit of 65%, as in severe polycythemia (too many RBCs).
**C (45%):** that is the **RBC layer** (the hematocrit itself), not plasma.
**E (75%):** would need a hematocrit of 25%, as in severe anemia.
**Exam twist:** The reverse is asked too (plasma 55% → hematocrit 45%), or "what makes up the buffy coat?" (leukocytes and platelets, under 1%).

**2.** *2026 TA review*
A tube of blood is allowed to clot and is then centrifuged. The supernatant fluid obtained is best described as:

**A.** Plasma containing coagulation factors
**B.** Serum lacking coagulation factors consumed during clotting
**C.** Whole blood without erythrocytes
**D.** Buffy coat fluid
**E.** Lymph

**Answer: B. Serum lacking coagulation factors consumed during clotting**

The key words are "**allowed to clot**." Fibrinogen turns into fibrin and traps the cells in the clot. The fluid left on top is **serum** = plasma **minus fibrinogen** and the used-up clotting factors.
**A (plasma with coagulation factors):** plasma comes from blood spun with an **anticoagulant**, so it keeps fibrinogen and the factors.
**C (whole blood without erythrocytes):** would still hold leukocytes and platelets; after clotting, all cells are in the clot.
**D (buffy coat fluid):** the buffy coat is a thin **cell layer** (leukocytes + platelets) in anticoagulated blood, not a fluid.
**E (lymph):** tissue fluid carried in lymphatic vessels; it does not come from a tube of blood.
**Exam twist:** Flipped version: "blood drawn into an anticoagulant tube and spun; the top layer is?" = **plasma** (55%, still contains fibrinogen).

### Erythrocytes

**3.** *2026 TA review*
A patient has erythrocytes with reduced membrane flexibility, making it difficult for them to pass through narrow capillaries. Which structural component normally supports the biconcave shape and flexibility of erythrocytes?

**A.** Myeloperoxidase in primary granules
**B.** Spectrin lattice anchored by ankyrin
**C.** Hemoglobin bound to fibrinogen
**D.** Marginal microtubule bundle
**E.** Histamine-containing granules

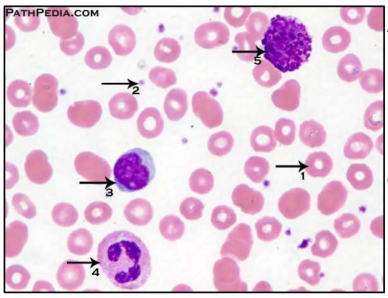
**Answer: B. Spectrin lattice anchored by ankyrin**

Under the RBC (red blood cell) membrane lies a mesh of **spectrin**, tied to the membrane by **ankyrin**. This cytoskeleton keeps the **biconcave disc** shape and lets the cell **bend** through capillaries narrower than its 7.5-micrometer width.
**A (myeloperoxidase in primary granules):** a killing enzyme in **neutrophil** azurophilic granules. RBCs have no granules.
**C (hemoglobin bound to fibrinogen):** hemoglobin carries oxygen inside the RBC; fibrinogen is a plasma clotting protein. They do not bind each other or shape the cell.
**D (marginal microtubule bundle):** keeps the disc shape of **platelets**.
**E (histamine-containing granules):** belong to **basophils and mast cells**.
**Exam twist:** Clinical version: **hereditary spherocytosis**. A spectrin or ankyrin defect makes round, rigid RBCs that get trapped and destroyed in the **spleen**.

## Leukocyte emigration and classification

**4.** *2026 blood session*
Identify the labeled formed elements in this peripheral smear

**Q1.** Identify the labeled formed elements in this peripheral smear




A. 1)RBC, 2)extruded nucleus, 3)lymphocyte, 4)eosinophil, 5)basophil

B. 1)RBC, 2)reticulocyte, 3)monocyte, 4)neutrophil, 5) basophil

C. 1)RBC, 2)platelet, 3)lymphocyte, 4)neutrophil, 5)basophil

D. 1)Reticulocyte, 2) platelet, 3)lymphocyte, 4)neutrophil, 5)basophil

**A.** 1)RBC, 2)extruded nucleus, 3)lymphocyte, 4)eosinophil, 5)basophil

**B.** 1)RBC, 2)reticulocyte, 3)monocyte, 4)neutrophil, 5) basophil

**C.** 1)RBC, 2)platelet, 3)lymphocyte, 4)neutrophil, 5)basophil

**D.** 1)Reticulocyte, 2) platelet, 3)lymphocyte, 4)neutrophil, 5)basophil

**Answer: C. 1)RBC, 2)platelet, 3)lymphocyte, 4)neutrophil, 5)basophil**

1 = pink disc, pale center, no nucleus: **RBC**. 2 = tiny purple speck: **platelet**. 3 = round dark nucleus, thin **sky-blue rim**: **lymphocyte**. 4 = **3–5 lobes**, pale lilac cytoplasm: **neutrophil**. 5 = coarse **blue-black granules** hiding the nucleus: **basophil**.
**A (2 extruded nucleus, 4 eosinophil):** extruded erythroblast nuclei are eaten in the marrow and never reach blood. Cell 4 lacks the large **orange-red** granules of an eosinophil.
**B (2 reticulocyte, 3 monocyte):** a reticulocyte is a full-size, bluish RBC, not a speck. Cell 3 lacks the kidney-shaped nucleus and abundant cytoplasm of a monocyte.
**D (1 reticulocyte):** cell 1 is a normal pink RBC with central pallor. A reticulocyte is slightly larger and bluish.
**Exam twist:** The same smear can ask "which element has no nucleus and lives about 10 days?" (platelet, 2) or "which is the rarest leukocyte?" (basophil, 5).

## Neutrophils

**5.** *2026 Canvas practice quiz*

Match each item to the correct description — First leukocyte to arrive at the site of infection, if the patient is female a barr body will be visible

**A.** Basophil

**B.** Eosinophil

**C.** Monocyte

**D.** Platelet

**E.** Lymphocyte

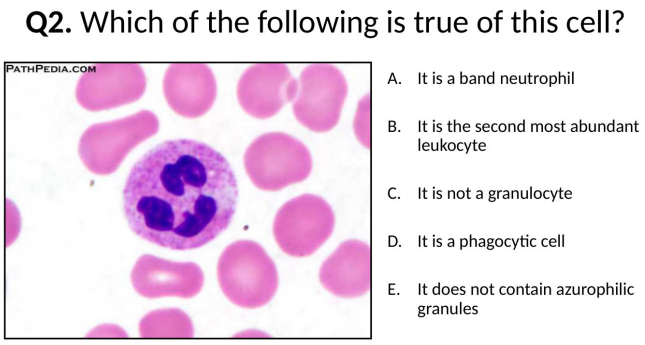
**F.** Neutrophil

**Answer: F. Neutrophil**

Neutrophils are the **first leukocytes** to reach an infection. In females the inactive X chromosome forms a **Barr body**; in a blood smear it shows as a small "**drumstick**" hanging off one lobe of the neutrophil nucleus.
**A (basophil):** rare (0–0.75%) allergy cell with histamine and heparin granules and IgE (immunoglobulin E) receptors. Its granules hide the nucleus.
**B (eosinophil):** bilobed nucleus with red-orange granules; fights parasitic worms and takes part in allergy.
**C (monocyte):** arrives **later** than neutrophils and becomes a tissue macrophage; kidney-shaped nucleus.
**D (platelet):** a cell fragment with **no nucleus**, so no Barr body; it works in clotting.
**E (lymphocyte):** round, unlobed nucleus; adaptive immunity, not the first responder.
**Exam twist:** The other matching prompts pair **MBP (major basic protein)** → eosinophil, **histamine + heparin** → basophil, and **precursor of macrophages** → monocyte.

**6.** *2026 blood session*

Which of the following is true of this cell?




**A.** It is a band neutrophil

**B.** It is the second most abundant leukocyte

**C.** It is not a granulocyte

**D.** It is a phagocytic cell

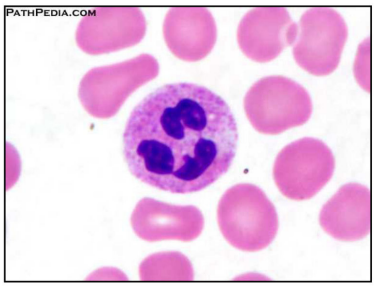
**E.** It does not contain azurophilic granules

**Answer: D. It is a phagocytic cell**

This is a **segmented neutrophil**: nucleus split into **3–5 lobes** joined by thin strands, pale lilac cytoplasm with fine granules, about twice the width of an RBC (red blood cell). Neutrophils are **active phagocytes** that eat and kill bacteria.
**A (band neutrophil):** a band has one unsegmented **horseshoe (C-shaped)** nucleus; this one has separate lobes.
**B (second most abundant):** neutrophils are the **most** abundant WBC (white blood cell), 54–62%. Lymphocytes are second (25–33%).
**C (not a granulocyte):** it is a granulocyte. The agranulocytes are lymphocytes and monocytes.
**E (no azurophilic granules):** it has **azurophilic (primary) granules** with MPO (myeloperoxidase), lysozyme and defensins, plus specific granules.
**Exam twist:** Expect "first leukocyte at a bacterial infection," "Barr body (drumstick) in a female," or "MPO in azurophilic granules"; all point to the neutrophil.

**7.** *2026 blood session*
Which of the following is true of this cell?

## **Q2.** Which of the following is true of this cell?




A. It is a band neutrophil
B. It is the second most abundant leukocyte
C. Transendothelial migration is facilitated by the elongated and segmented nucleus
D. It is a weakly phagocytic cell
E. It contains azurophilic, but not specific granules

**A.** It is a band neutrophil
**B.** It is the second most abundant leukocyte
**C.** Transendothelial migration is facilitated by the elongated and segmented nucleus
**D.** It is a weakly phagocytic cell
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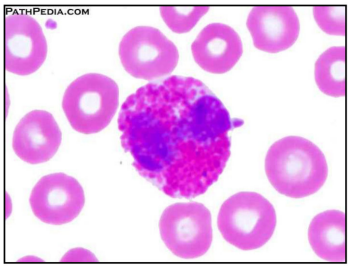
**Answer: C. Transendothelial migration is facilitated by the elongated and segmented nucleus**

Same picture: a **segmented neutrophil** with **3–5 thin lobes** and pale lilac cytoplasm. A thin, lobed nucleus can **squeeze between endothelial cells** during diapedesis (leaving the blood through the wall of a postcapillary venule).
**A (band neutrophil):** a band has one unsegmented horseshoe nucleus; this nucleus is pinched into lobes.
**B (second most abundant):** neutrophils are the **most** abundant WBC (white blood cell), 54–62%. Lymphocytes are second.
**D (weakly phagocytic):** neutrophils are **avid** phagocytes of bacteria. "Weakly" is the trap word.
**E (azurophilic but not specific granules):** mature neutrophils have **both**. Only azurophilic granules = the promyelocyte stage; specific granules first appear in the myelocyte.
**Exam twist:** The emigration steps are a favorite: **selectins** → rolling, then integrins bind ICAM-1 (intercellular adhesion molecule 1) → firm adhesion, then **diapedesis** at **postcapillary venules**.

## Eosinophils and basophils

**8.** *2026 blood session*
Which of the following is true of this cell?

**Q3.** Which of the following is true of this cell?




A. Its main function is to destroy viruses
B. It is a band form
C. It is not involved in allergic reactions
D. The granules contain major basic protein
E. It is almost as abundant as neutrophils

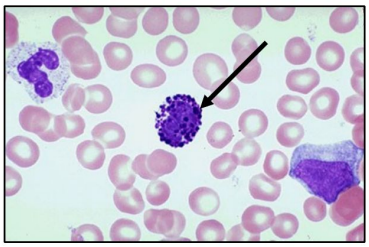
**A.** Its main function is to destroy viruses
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**C.** It is not involved in allergic reactions
**D.** The granules contain major basic protein
**E.** It is almost as abundant as neutrophils

**Answer: D. The granules contain major basic protein**

Look for a **bilobed nucleus** and cytoplasm packed with **large red-orange granules** = **eosinophil**. Its specific granules hold **MBP (major basic protein)**, about half the granule protein; MBP is what makes the granules stain red with eosin. They also hold eosinophil peroxidase.
**A (destroys viruses):** killing virus-infected cells is the job of NK (natural killer) cells and CD8 (cluster of differentiation 8) cytotoxic T cells. Eosinophils kill **parasitic worms**.
**B (band form):** a band is a young neutrophil with a horseshoe nucleus and pale lilac granules.
**C (not in allergy):** false. Eosinophils rise in **allergic** inflammation such as asthma.
**E (almost as abundant as neutrophils):** eosinophils are only **1–3%** of WBCs (white blood cells); neutrophils are 54–62%.
**Exam twist:** A vignette with a worm infection (such as schistosomiasis) or asthma plus a high eosinophil count points to this cell and MBP.

**9.** *2025 faculty blood session*
Q5: What are the biologically active molecules contained within the granules of the cell indicated by an arrow?

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A. Histamine and DNase
B. Histamine and ribosomal RNA
C. Histamine and heparin
D. Lysosomes and hydrogen peroxide
E. Calcium and lipase

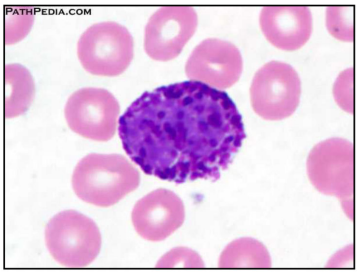
**A.** Histamine and DNase
**B.** Histamine and ribosomal RNA
**C.** Histamine and heparin
**D.** Lysosomes and hydrogen peroxide
**E.** Calcium and lipase

**Answer: C. Histamine and heparin**

The arrow points to a small cell whose nucleus is hidden by **coarse blue-black granules** = **basophil**. Its specific granules hold **histamine and heparin** (plus eosinophil chemotactic factor). Heparin is a strongly acidic GAG (glycosaminoglycan), which is why the granules take up the basic dye so darkly.
**A (histamine + DNase):** DNase (an enzyme that breaks down genetic material) is not a basophil mediator; only histamine fits.
**B (histamine + ribosomal RNA):** ribosomal RNA (ribonucleic acid) is the blue in reticulocytes and young blast cells, not granule cargo.
**D (lysosomes + hydrogen peroxide):** neutrophil killing tools. The neutrophil is the lobed cell at the left edge.
**E (calcium + lipase):** calcium is stored in platelets; lipase is a pancreatic digestive enzyme.
**Exam twist:** Granule matching is the classic version: **MPO (myeloperoxidase)** → neutrophil, **MBP (major basic protein)** → eosinophil, **histamine + heparin** → basophil and mast cell.

**10.** *2026 blood session*
Which of the following is true of this cell?

## **Q4.** Which of the following is true of this cell?




A. Its main function is phagocytosis
B. Its main purpose is to respond to produce antibodies
C. It performs functions similar to the mast cell
D. Its granules do not contain histamine
E. Its nucleus is not lobulated

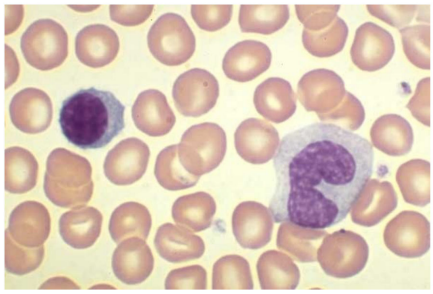
**A.** Its main function is phagocytosis
**B.** Its main purpose is to respond to produce antibodies
**C.** It performs functions similar to the mast cell
**D.** Its granules do not contain histamine
**E.** Its nucleus is not lobulated

**Answer: C. It performs functions similar to the mast cell**

Large **dark purple-black granules** cover and hide the nucleus = **basophil**. Like a mast cell, it has **IgE (immunoglobulin E) receptors** and granules of **histamine and heparin**, so it drives allergic reactions and supplements mast cells in tissue.
**A (main function phagocytosis):** that is the neutrophil and the monocyte/macrophage.
**B (makes antibodies):** antibodies come from **plasma cells** (activated B lymphocytes).
**D (no histamine):** false. Histamine is a main basophil mediator.
**E (nucleus not lobulated):** the nucleus is **bilobed or S-shaped**; it is just hidden under the granules.
**Exam twist:** Asked as "rarest leukocyte (0–0.75%), often absent from a slide" or "which blood cell binds IgE and releases histamine in allergy"; both are the basophil.

## Lymphocytes and monocytes

**11.** *2026 blood lab deck*
The leukocyte on the left is a…

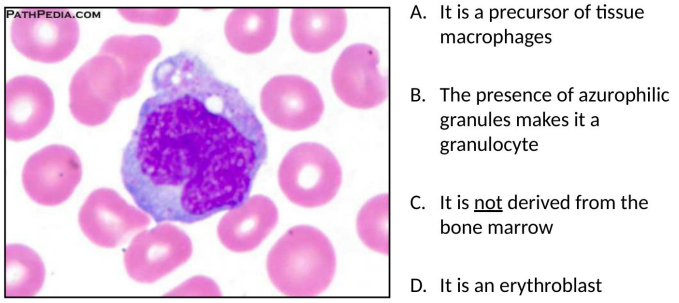



**A.** Neutrophil
**B.** Eosinophil
**C.** Basophil
**D.** Monocyte
**E.** Lymphocyte

**Answer: E. Lymphocyte**

The left cell is about the **size of an RBC** (red blood cell), with a **round, dense, dark nucleus** that fills almost the whole cell and only a **thin rim of pale blue cytoplasm** = **small lymphocyte**.
**A (neutrophil):** would show 3–5 nuclear lobes in pale lilac cytoplasm.
**B (eosinophil):** would show a bilobed nucleus and bright red-orange granules.
**C (basophil):** dark purple-black granules would hide the nucleus.
**D (monocyte):** that is the **larger cell on the right**: kidney-shaped nucleus with lacy chromatin and abundant grey-blue cytoplasm.
**Exam twist:** The same pair is asked as "the cell on the right is a…" (monocyte, the largest leukocyte) or "the second most abundant leukocyte" (lymphocyte, 25–33%).

**12.** *2026 blood session*
Which of the following is true of this cell?



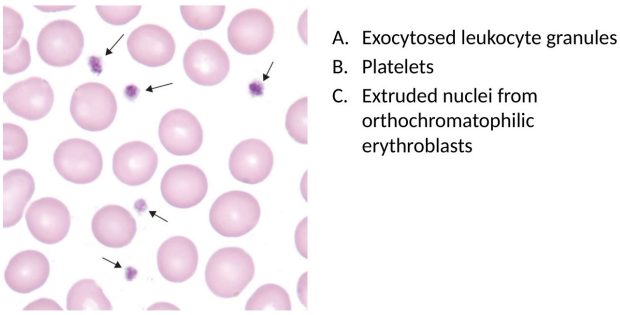

**A.** It is a precursor of tissue macrophages
**B.** The presence of azurophilic granules makes it a granulocyte
**C.** It is not derived from the bone marrow
**D.** It is an erythroblast

**Answer: A. It is a precursor of tissue macrophages**

Large cell with an **indented (kidney-shaped) nucleus**, **lacy, pale chromatin**, and plenty of **grey-blue cytoplasm** with small vacuoles = **monocyte**. It leaves the blood and becomes a **tissue macrophage**.
**B (azurophilic granules make it a granulocyte):** its azurophilic granules are just lysosomes. It has no specific granules, so it is an **agranulocyte**.
**C (not from bone marrow):** monocytes are made in the **bone marrow** (monoblast → promonocyte → monocyte).
**D (erythroblast):** erythroblasts stay in the marrow and have a **round**, dense nucleus with blue-to-pink cytoplasm and no vacuoles.
**Exam twist:** Monocytes also give rise to **osteoclasts and microglia** (the mononuclear phagocyte system); "which blood cell becomes an osteoclast?" = monocyte.

## Platelet structure: LM and EM

**13.** *2026 blood lab deck*
The arrows point towards which of the following?




**A.** Exocytosed leukocyte granules

**B.** Platelets

**C.** Extruded nuclei from orthochromatophilic erythroblasts

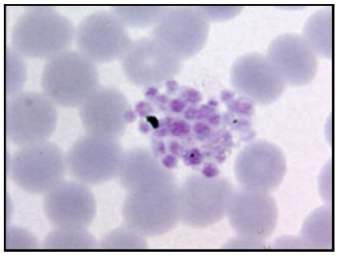
**Answer: B. Platelets**

The arrows mark tiny (2–3 micrometer) **purple fragments with no nucleus**, far smaller than the RBCs (red blood cells). Each has a **dark granular center (granulomere)** and a **pale rim (hyalomere)** = **platelets**.
**A (exocytosed leukocyte granules):** free granules would be scattered dots well under 1 micrometer, not discrete bodies with a pale rim.
**C (extruded nuclei of orthochromatophilic erythroblasts):** these are eaten by marrow macrophages and never reach normal blood. They would also be round and uniformly dark, with no pale rim.
**Exam twist:** Asked as "which formed element lives about 10 days and comes from megakaryocytes?" or as an EM (electron microscope) question: alpha granules (clotting factors, PDGF = platelet-derived growth factor) vs dense granules (ADP = adenosine diphosphate, serotonin).

## Platelet origin and hemostasis

**14.** *2025 faculty blood session*
Which of the following is true of the central small structures in this blood smear?

**Q9.** Which of the following is true of the central small structures in this blood smear?



A. Electron microscopy shows them to be perfectly spherical

B. They have a peripheral granulomere and a central hyalomere

C. They have a minute central nucleus

D. They are agranular

E. They are derived from polyploid bone marrow cells

**A.** Electron microscopy shows them to be perfectly spherical
**B.** They have a peripheral granulomere and a central hyalomere
**C.** They have a minute central nucleus
**D.** They are agranular
**E.** They are derived from polyploid bone marrow cells

**Answer: E. They are derived from polyploid bone marrow cells**

The central clump of small purple fragments, much smaller than the RBCs (red blood cells), is **platelets**; platelets **clump** in smears. They are pinched off from **megakaryocytes**, giant marrow cells with one lobed **polyploid** nucleus (up to 64N, meaning 64 single chromosome sets, over 30 times the normal amount of DNA, deoxyribonucleic acid).
**A (perfectly spherical):** EM (electron microscopy) shows **flat discs**, held in shape by a marginal bundle of microtubules.
**B (peripheral granulomere, central hyalomere): reversed**. The pale **hyalomere** is the rim; the dark granular **granulomere** is the center.
**C (minute central nucleus):** platelets are cytoplasmic fragments with **no nucleus**; the dark center is granules.
**D (agranular):** they have **alpha granules** (clotting factors, growth factor) and **dense (delta) granules** (ADP = adenosine diphosphate, serotonin).
**Exam twist:** Also asked as "how do platelets reach the blood?": megakaryocytes push **proplatelets** through the marrow **sinusoid** wall, and platelets break off the tips.

# Topic 9B · Hematopoiesis & bone marrow (H11B)

## Hematopoietic sites by age; marrow conversion

**15.** *2026 TA review*

Question: A bone marrow biopsy shows hematopoietic tissue replaced by adipose tissue in most long bones. This finding represents:

**A.** Normal age-related replacement of red marrow by yellow marrow

**B.** Pathologic aplastic anemia

**C.** Myelophthisic anemia

**D.** Bone marrow fibrosis

**E.** Iron deficiency anemia

**Answer: A. Normal age-related replacement of red marrow by yellow marrow**

Red marrow fills all bones at birth. From childhood on, **fat (yellow marrow)** replaces it, starting in the **shafts of long bones**. In adults (over about 25), red marrow stays mainly in the **axial skeleton** (vertebrae, sternum, ribs, pelvis) and the ends of long bones. Fatty long bones = **normal**.

**B (aplastic anemia):** fat replaces marrow **everywhere**, axial skeleton included, with low counts of all blood cells.

**C (myelophthisic anemia):** marrow is crowded out by **tumor** or other invading tissue, not fat.

**D (marrow fibrosis):** marrow is replaced by **fibrous scar**; the aspirate often comes back dry.

**E (iron deficiency anemia):** marrow stays cellular; only the stored iron is missing on Prussian blue stain.

**Exam twist:** Asked as "where do you biopsy an adult's marrow?" (**posterior iliac crest**, because red marrow persists in the axial skeleton) or "fetal blood-forming sites in order" (yolk sac → liver/spleen → marrow).

## Red marrow histology, megakaryocytes, sinusoids (incl. IDs)

**16.** *2026 TA review*

How do bone marrow cells enter circulation?

**A.** Passing through the discontinuous sinusoidal endothelium

**B.** Passing through the fenestrated endothelium

**C.** Passing through the continuous endothelium

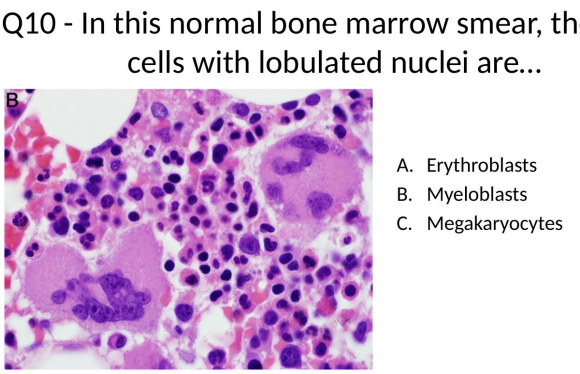
**D.** Passing through the tight junctions

**Answer: A. Passing through the discontinuous sinusoidal endothelium**

Marrow blood vessels are wide **sinusoids** with a **discontinuous** lining: openings in the endothelium and a broken basal lamina. New blood cells squeeze through these openings into the lumen; megakaryocytes push proplatelets in the same way.
**B (fenestrated endothelium):** small pores (gut, endocrine glands, kidney) that pass fluid and small molecules, not whole cells.
**C (continuous endothelium):** unbroken lining with a full basal lamina (muscle, lung, brain); cells cannot pass.
**D (tight junctions):** seal neighboring cells together; nothing passes through them.
**Exam twist:** The same vessel type appears in the **liver** and **spleen**; in the spleen, blood cells must squeeze between **stave cells** to re-enter the sinusoids.

**17.** *2026 blood lab deck*

In this normal bone marrow smear, the large cells with lobulated nuclei are…




**A.** Erythroblasts

**B.** Myeloblasts

**C.** Megakaryocytes

**Answer: C. Megakaryocytes**

The two **giant cells**, several times larger than anything around them, have broad pale-pink granular cytoplasm and **one large multi-lobed nucleus** = **megakaryocytes**, the platelet makers. The nucleus is **polyploid** (up to 64N, meaning 64 single chromosome sets) because it copies its DNA (deoxyribonucleic acid) without dividing.
**A (erythroblasts):** small cells with a **round, very dark nucleus**; many sit in the background.
**B (myeloblasts):** medium cells with **one round nucleus**, fine chromatin and nucleoli; never giant or lobulated.
**Exam twist:** Asked as "what does this giant marrow cell make?" (**platelets**, via proplatelets pushed into sinusoids). Don't confuse it with an osteoclast, which has many separate nuclei and sits on bone.

## Erythropoiesis stages and reticulocytes (incl. IDs)

**18.** *2026 blood session · 2025 faculty blood session*

In order of development, what are the 6 stages of erythropoiesis?

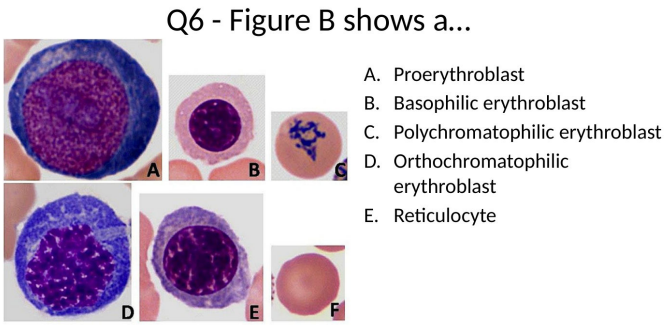
- **A.** Proerythroblast, basophilic erythroblast, polychromatophilic erythroblast, orthochromatophilic erythroblast, reticulocyte, erythrocyte
- **B.** Erythroblast, basophilic erythroblast, polychromatophilic erythroblast, orthochromatophilic erythroblast, reticulocyte, erythrocyte
- **C.** Proerythroblast, orthochromatophilic erythroblast, basophilic erythroblast, polychromatophilic erythroblast, reticulocyte, erythrocyte
- **D.** Proerythroblast, polychromatophilic erythroblast, basophilic erythroblast, orthochromatophilic erythroblast, reticulocyte, erythrocyte

**Answer: A. Proerythroblast, basophilic erythroblast, polychromatophilic erythroblast, orthochromatophilic erythroblast, reticulocyte, erythrocyte**

Follow the **cytoplasm color**: deep **blue** (ribosomes) → **mixed grey-lilac** (ribosomes + hemoglobin) → **pink** (hemoglobin). Meanwhile the cell shrinks, the nucleus condenses, and the orthochromatophilic erythroblast **extrudes the nucleus**.
**B (starts with "erythroblast"):** the first named stage is the **pro**erythroblast. "Erythroblast" alone is not a stage.
**C (orthochromatophilic second):** puts the pink, nearly mature stage before the blue ones. Color must go blue → mixed → pink.
**D (polychromatophilic before basophilic):** a mixed color cannot come before pure blue, because hemoglobin builds up gradually.
**Exam twist:** Single-stage versions: "most basophilic cytoplasm" = **basophilic erythroblast**; "extrudes the nucleus" = **orthochromatophilic erythroblast**; "no nucleus but still has polyribosomes" = **reticulocyte**.

**19.** *2026 blood lab deck*

Figure B shows a…



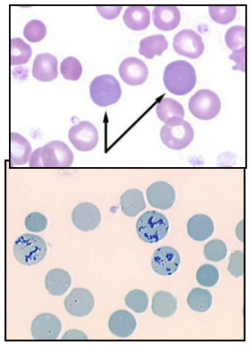

- **A.** Proerythroblast
- **B.** Basophilic erythroblast
- **C.** Polychromatophilic erythroblast
- **D.** Orthochromatophilic erythroblast
- **E.** Reticulocyte

**Answer: D. Orthochromatophilic erythroblast**

Figure B is **small**, its cytoplasm is **pink** like a mature RBC (red blood cell) because it is full of hemoglobin, and it still has a **small, dense, round nucleus**. Pink cytoplasm + a nucleus = the **last nucleated stage**, the one that extrudes its nucleus.
**A (proerythroblast = Figure A):** the **largest** cell; fine chromatin with nucleoli and a rim of deep-blue cytoplasm.
**B (basophilic erythroblast = Figure D):** large, **intensely blue** cytoplasm, coarser chromatin.
**C (polychromatophilic erythroblast = Figure E): grey-lilac** (blue + pink mixed) cytoplasm, checkerboard chromatin.
**E (reticulocyte = Figure C): no nucleus**; the dark blue squiggles are RNA (ribonucleic acid) on a supravital stain. Figure F is the mature RBC.
**Exam twist:** The same panel returns for every letter, or as "which stage **extrudes its nucleus**?" (Figure B) and "which stage has the most basophilic cytoplasm?" (Figure D).

**20.** *2026 blood session*
Which of the following is true of the cells at the tips of the arrows?

**Q8.** Which of the following is true of the cells at the tips of the arrows?



A. They are erythroblasts

B. They have excessive DNA

C. Their numbers decrease when there is blood loss

D. They are the same cell type as those in the bottom picture containing bluish material

E. They are infected with malaria organisms

**A.** They are erythroblasts

**B.** They have excessive DNA

**C.** Their numbers decrease when there is blood loss

**D.** They are the same cell type as those in the bottom picture containing bluish material

**E.** They are infected with malaria organisms

**Answer: D. They are the same cell type as those in the bottom picture containing bluish material**

Top (Wright stain): the arrowed cells have **no nucleus**, are **slightly larger** than RBCs (red blood cells), and are faintly **bluish-purple** = **reticulocytes**. Bottom (supravital stain such as new methylene blue): their leftover **ribosomal RNA** (ribonucleic acid) clumps into a **blue net (reticulum)**. Same cells, two stains.
**A (erythroblasts):** erythroblasts have a **nucleus** and normally stay in the marrow.
**B (excessive DNA):** no nucleus, so no DNA (deoxyribonucleic acid). The blue is RNA.
**C (decrease with blood loss):** blood loss or hemolysis makes the marrow release **more** reticulocytes.
**E (malaria):** malaria parasites look like small **rings with a dark chromatin dot** inside RBCs, not a diffuse bluish tint or a net.
**Exam twist:** "Which test shows the marrow is responding to anemia?" = **reticulocyte count** (normally about 1–2% of RBCs; ↑ after bleeding or hemolysis).

## Granulopoiesis; erythropoiesis vs granulopoiesis (incl. IDs)

**21.** *2026 blood session*

In order of development, what are the 6 stages of neutrophil granulopoiesis?

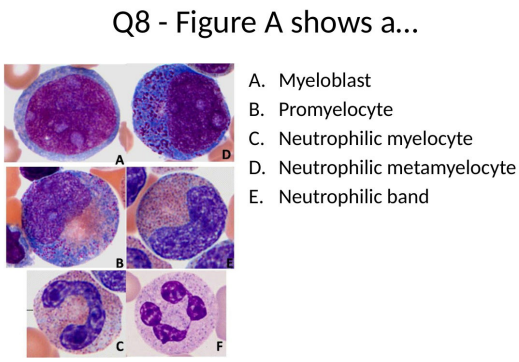
- **A.** Myeloblast, promyelocyte, myelocyte, metamyelocyte, band, neutrophil
- **B.** Promyelocyte, myeloblast, myelocyte, metamyelocyte, band, neutrophil
- **C.** Myelocyte, myeloblast, promyelocyte, metamyelocyte, band, neutrophil
- **D.** Metamyelocyte, myeloblast, promyelocyte, myelocyte, band, neutrophil
- **E.** Band, myeloblast, promyelocyte, myelocyte, metamyelocyte, neutrophil

**Answer: A. Myeloblast, promyelocyte, myelocyte, metamyelocyte, band, neutrophil**

Track two clues. **Granules:** none in the myeloblast, **azurophilic** granules in the promyelocyte, **specific granules** first in the myelocyte. **Nucleus:** round → indented (**kidney**, metamyelocyte) → **horseshoe** (band) → **3–5 lobes** (segmented).
**B (promyelocyte first):** the granule-free **myeloblast** must come before the granule-filled promyelocyte.
**C (myelocyte first):** the myelocyte already has specific granules, so it comes after the blast and promyelocyte.
**D (metamyelocyte first):** a kidney-shaped nucleus is a late feature; it follows the myelocyte.
**E (band first):** the band is the last stage before segmentation, not the start.
**Exam twist:** Single-stage versions: "first stage with **specific granules**" = **myelocyte**; "most common immature granulocyte in marrow" = myelocyte; "left shift" in infection = more **bands** in the blood.

**22.** *2026 blood lab deck*

Figure A shows a…

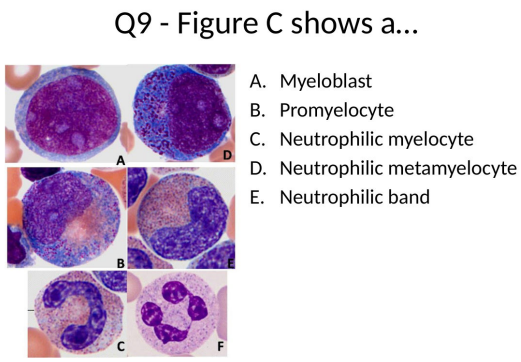



- **A.** Myeloblast
- **B.** Promyelocyte
- **C.** Neutrophilic myelocyte
- **D.** Neutrophilic metamyelocyte
- **E.** Neutrophilic band

**Answer: A. Myeloblast**

Figure A is a large cell with a **big round nucleus**, **fine, even chromatin** and **several pale nucleoli**, plus a thin rim of blue cytoplasm with **no granules**. A granule-free blast in the granulocyte panel = **myeloblast**.
**B (promyelocyte = Figure D):** similar size, but the cytoplasm is **packed with dark azurophilic granules**.
**C (neutrophilic myelocyte = Figure B):** round-oval off-center nucleus, a pale area beside it, and fine **specific granules**.
**D (neutrophilic metamyelocyte = Figure E):** nucleus **indented** like a kidney bean.
**E (neutrophilic band = Figure C): horseshoe (C-shaped)** nucleus of even width. Figure F is the mature segmented neutrophil.
**Exam twist:** Expect the same panel with a different letter, or "the first stage with specific granules" (myelocyte, Figure B).

**23.** *2026 blood lab deck*
Figure C shows a…

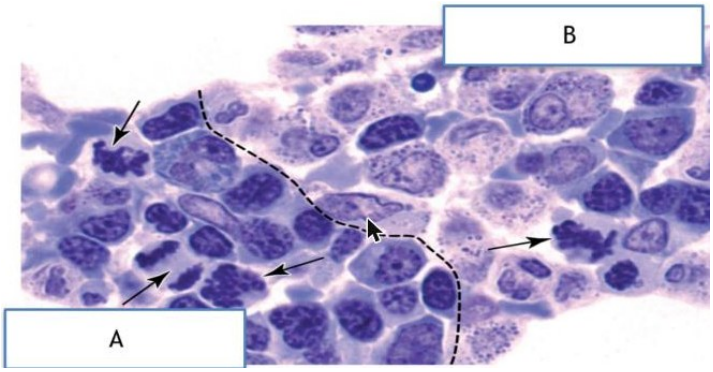



**A.** Myeloblast
**B.** Promyelocyte
**C.** Neutrophilic myelocyte
**D.** Neutrophilic metamyelocyte
**E.** Neutrophilic band

**Answer: E. Neutrophilic band**

Figure C has a **horseshoe (C-shaped) nucleus** of fairly even width, curved around the cell but **not pinched into separate lobes**, in pale cytoplasm with fine granules = **band** neutrophil.
**A (myeloblast = Figure A):** round nucleus, fine chromatin, nucleoli, **no granules**.
**B (promyelocyte = Figure D):** round nucleus with cytoplasm full of **dark azurophilic granules**.
**C (neutrophilic myelocyte = Figure B): round-oval** nucleus; specific granules just starting.
**D (neutrophilic metamyelocyte = Figure E):** nucleus only **indented** (kidney-shaped) and still broad. Figure F, with 3–5 lobes joined by thin strands, is the mature segmented neutrophil.
**Exam twist:** Clinical version: in an acute bacterial infection the marrow releases young neutrophils early, so **bands rise** in the blood ("left shift").

**24.** *2026 Canvas practice quiz*
In the image below we observe a bone marrow micrograph. "A" is labeling the side of the image where ____ is predominantly seen.




**A.** thrombopoiesis
**B.** erythropoiesis
**C.** granulopoiesis

**Answer: B. erythropoiesis**

Side A (lower left of the dashed line) has **tightly packed cells** with **round, dark nuclei** of coarse chromatin and smooth, **granule-free** cytoplasm = **erythroid** cells (erythroblasts). The arrows mark cells in **mitosis**.
**A (thrombopoiesis):** would show **giant megakaryocytes** with lobed nuclei; none are here.
**C (granulopoiesis):** that is **side B**: paler, larger cells with **speckled, granular cytoplasm** and oval to indented (kidney-shaped) nuclei.
**Exam twist:** Expect "which side shows granulopoiesis?" (B), or a marrow section asking you to name the giant cell (megakaryocyte) or the large empty spaces (**adipocytes**).

## Lab clinical correlates: anemia, inflammation

**25.** *2025 faculty lab deck*

Case No. 5: In an anemia workup, which of the following should be ordered first to test for effective erythropoiesis?

**A.** Bone marrow aspirate

**B.** Bone marrow trephine biopsy

**C.** Red cell survival study

**D.** Reticulocyte count

**E.** Serum bilirubin

**Answer: D. Reticulocyte count**

Reticulocytes are the newest RBCs (red blood cells) released from marrow. Counting them is the **first, simple blood test** of whether the marrow is producing well. **High** = marrow responding (bleeding, hemolysis); **low** = marrow not keeping up (iron deficiency, marrow failure).
**A (bone marrow aspirate):** invasive; done later if the cause is still unclear.
**B (trephine biopsy):** a core of marrow from the iliac crest; more invasive still, also a later step.
**C (red cell survival study):** measures how long RBCs **last** (destruction), not how fast they are made.
**E (serum bilirubin):** bilirubin rises from hemoglobin breakdown, so it marks **hemolysis**, not production.
**Exam twist:** Paired with the stain question: reticulocytes show a **blue net** on a **supravital stain** (new methylene blue or brilliant cresyl blue), not on Wright stain.

**26.** *2026 TA review*

A 19 year old male presents to the emergency department with severe pain in his chest and extremities. He reports multiple similar episodes, particularly during periods of dehydration and physical exertion. A peripheral blood smear reveals numerous elongated, crescent-shaped erythrocytes. Which of the following mechanisms most directly explains the abnormal erythrocyte morphology?

**A.** Decreased production of hemoglobin due to iron deficiency

**B.** Polymerization of abnormal hemoglobin S under low-oxygen conditions

**C.** Autoimmune destruction of circulating erythrocytes

**D.** Failure of erythrocytes to extrude their nuclei during maturation

**E.** Increasing production of erythrocytes due to a JAK2 mutation

**Answer: B. Polymerization of abnormal hemoglobin S under low-oxygen conditions**

Crescent-shaped RBCs (red blood cells) + pain crises with dehydration and exertion = **sickle cell anemia**. A point mutation makes **HbS (hemoglobin S)**, which **polymerizes at low oxygen** and bends the cell into a rigid sickle that blocks small vessels.
**A (iron deficiency):** makes **small, pale** RBCs, not sickles.
**C (autoimmune destruction):** antibody-coated RBCs lose membrane and become **spherocytes** (small, round, no central pallor).
**D (failure to extrude nuclei):** would leave **nucleated RBCs** in the blood. Nuclear extrusion happens at the orthochromatophilic erythroblast.
**E (JAK2 mutation):** a JAK2 (Janus kinase 2) mutation causes polycythemia vera: **too many** RBCs of normal shape.
**Exam twist:** Rigid sickle cells cannot squeeze between splenic stave cells, so **splenic macrophages** destroy them → hemolytic anemia with a **high reticulocyte count**.

# Topic 10 · Lymphoid cells, tissues & organs (H14)

## Lymphoid organs and lymph circulation

**27.** *2026 TA review*

Which statement best describes the defining functional distinction between primary (central) and secondary (peripheral) lymphoid organs?

**A.** Primary organs filter lymph, while secondary organs filter blood.

**B.** Primary organs contain encapsulated nodules, while secondary organs are unencapsulated.

**C.** Primary organs are where lymphocytes form and mature, while secondary organs are where immune reactions take place.

**D.** Primary organs rely on high endothelial venules (HEVs) for cell entry, while secondary organs do not.

**Answer: C. Primary organs are where lymphocytes form and mature, while secondary organs are where immune reactions take place.**

**Primary (central)** organs are the **bone marrow** (B cells) and **thymus** (T cells), where lymphocytes are made and trained. **Secondary (peripheral)** organs are the **lymph nodes, spleen and MALT** (mucosa-associated lymphoid tissue), where B cells, T cells and antigen-presenting cells meet antigen.

**A (primary filter lymph, secondary filter blood):** nodes filter lymph and the spleen filters blood, and **both are secondary**.

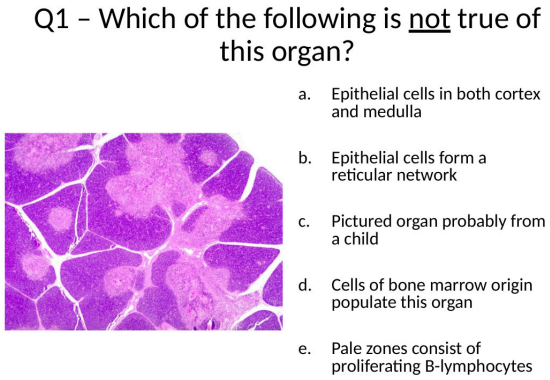
**B (encapsulated vs unencapsulated):** not the defining split. The thymus (primary) and lymph nodes (secondary) both have capsules; MALT has none.

**D (HEVs in primary organs only):** reversed. HEVs (high endothelial venules, where blood lymphocytes enter) are in **secondary** organs such as lymph nodes and Peyer's patches.

**Exam twist:** Expect "which is a primary lymphoid organ?" with lymph node, spleen and tonsil as distractors; the answer is **thymus** or **bone marrow**.

## Thymus: architecture, age changes, identification

**28.** *2026 lymphoid session*
Which of the following is not true of this organ?

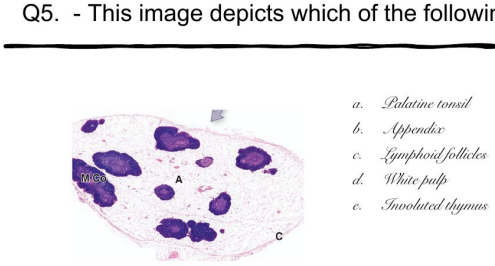



**A.** Epithelial cells in both cortex and medulla
**B.** Epithelial cells form a reticular network
**C.** Pictured organ probably from a child
**D.** Cells of bone marrow origin populate this organ
**E.** Pale zones consist of proliferating B-lymphocytes

**Answer: E. Pale zones consist of proliferating B-lymphocytes**

Lobules split by thin septa, each with a **dark outer cortex** and a **pale medulla**, no nodules, and almost no fat = **child's thymus**. The pale zones are **medulla** (fewer, larger T cells, epithelial cells, Hassall corpuscles). The thymus has **no B-cell germinal centers**.
**A (epithelial cells in cortex and medulla):** true. TECs (thymic epithelial cells) sit in both zones.
**B (epithelial cells form a reticular network):** true. TECs join by desmosomes into a **cytoreticulum** that replaces reticular fibers.
**C (probably a child):** true. Dense lymphoid lobules without fatty involution.
**D (bone marrow cells populate it):** true. T-cell progenitors come from the **marrow** through the blood.
**Exam twist:** The lab asks "does this thymus contain germinal centers?" Answer: **no**. At low power the pale medulla can pass for germinal centers.

**29.** *past faculty session*
Q5. - This image depicts which of the following?



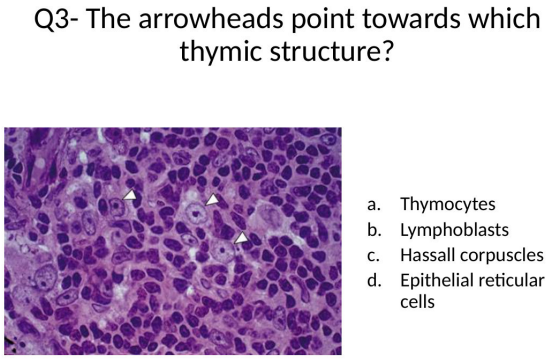

**A.** Palatine tonsil
**B.** Appendix
**C.** Lymphoid follicles
**D.** White pulp
**E.** Involuted thymus

**Answer: E. Involuted thymus**

Most of the section is **adipose tissue (A)** inside a thin **capsule (C)**. Small islands of dark lymphoid tissue keep a **cortex (Co) and medulla (M)** pattern = **adult involuted thymus**. The thymus is largest at puberty and is then replaced by fat.
**A (palatine tonsil):** would show **stratified squamous epithelium** dipping into deep crypts, with nodules along them.
**B (appendix):** would show a gut wall: central lumen, glands with goblet cells, muscle layers, nodules ringing the mucosa.
**C (lymphoid follicles):** follicles sit inside lymphoid organs or mucosa, not as islands in fat; these islands have cortex and medulla, not germinal centers.
**D (white pulp):** splenic white pulp sits in **red pulp** around a central arteriole, not in fat.
**Exam twist:** A child-vs-adult pair: the child thymus is dense lobules with no fat; the adult is fat with lymphoid islands. Both still have **Hassall corpuscles** in the medulla.

## Thymic epithelial cells and Hassall corpuscles

**30.** *2026 lymphoid lab deck*
The arrowheads point towards which thymic structure?




**A.** Thymocytes
**B.** Lymphoblasts
**C.** Hassall corpuscles
**D.** Epithelial reticular cells

**Answer: D. Epithelial reticular cells**

Among many small dark thymocytes, the arrowheads mark **large cells** with **pale, oval, euchromatic nuclei**, a **clear nucleolus**, and pale pink cytoplasm = **epithelial reticular cells** (TECs, thymic epithelial cells). They form the thymus's supporting meshwork and train developing T cells.
**A (thymocytes):** the **small, round, dense-dark** nuclei packed everywhere (developing T cells).
**B (lymphoblasts):** large dividing lymphocytes with a **deep blue** cytoplasm rim, found mainly in the outer cortex under the capsule; not pale cells with pink cytoplasm.
**C (Hassall corpuscles): pink concentric whorls** of flattened TECs in the medulla, not single nuclei.
**Exam twist:** A **cytokeratin** stain lights up TECs brown (they are true epithelium joined by desmosomes); also asked as "positive selection happens on **cortical** TECs".

**31.** *2026 TA review*
A thymic epithelial cell contributes to a barrier that prevents unregulated exposure of developing thymocytes to circulating antigens. Which combination correctly identifies the components of this barrier?

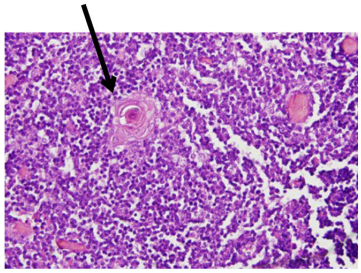
**A.** Capillary wall, perivascular connective tissue, and thymic epithelial cells
**B.** Capillary wall, follicular dendritic cells, and reticular fibers
**C.** High endothelial venules, macrophages, and medullary cords
**D.** Reticular fibers, endothelial stave cells, and plasma cells

**Answer: A. Capillary wall, perivascular connective tissue, and thymic epithelial cells**

The **blood–thymus barrier** in the **cortex** keeps blood antigens away from developing T cells. Layers from lumen outward: **capillary wall** (continuous endothelium, basal lamina, pericytes) → **perivascular connective tissue** with **macrophages** that catch leaked antigen → **thymic epithelial cells** with their basal lamina.
**B (follicular dendritic cells, reticular fibers):** follicular dendritic cells hold antigen in B-cell follicles. The thymus has **no follicles and no reticular fibers**.
**C (HEVs, macrophages, medullary cords):** HEVs (high endothelial venules) and medullary cords are **lymph node** structures.
**D (reticular fibers, stave cells, plasma cells):** stave cells line **splenic sinusoids**; reticular fibers and plasma cells fit spleen or node, not thymus.
**Exam twist:** Expect "the blood–thymus barrier is in the **cortex**, not the medulla" or "which barrier cell is a phagocyte?" (the perivascular **macrophage**).

**32.** *2026 lymphoid lab deck*
Name the indicated thymic structure

Q4- Name the indicated thymic structure



a. Squamous pearl
b. Malpighian corpuscle
c. Hassall corpuscle

**A.** Squamous pearl

**B.** Malpighian corpuscle

**C.** Hassall corpuscle

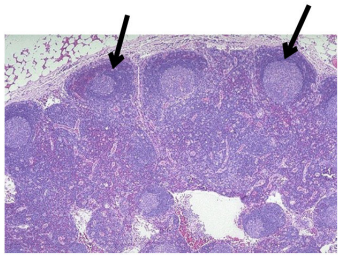
**Answer: C. Hassall corpuscle**

The arrow points to a **pink, concentric, onion-like whorl** of flattened cells, sometimes keratinized in the center, sitting in pale lymphoid tissue = **Hassall (thymic) corpuscle**. It is built from medullary TECs (thymic epithelial cells) and is found **only in the thymic medulla**.
**A (squamous pearl):** a keratin whorl in **squamous cell carcinoma**, a tumor, not normal lymphoid tissue.
**B (Malpighian corpuscle):** an old name for a **splenic white pulp nodule** (or a kidney glomerulus with its capsule); a round lymphoid nodule with an arteriole, not a pink whorl.
**Exam twist:** "Which organ?": one Hassall corpuscle proves **thymus**. Also "where are Hassall corpuscles?" = **medulla**, where negative selection and regulatory T-cell development happen.

## Lymph node organization and identification

**33.** *2026 lymphoid lab deck*
What lymph node region is identified by the arrows?

Q6- What lymph node region is identified by the arrows?



a. Cortex
b. Paracortex
c. Medulla

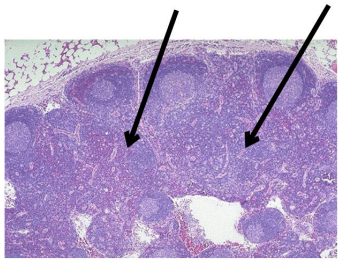
**A.** Cortex
**B.** Paracortex
**C.** Medulla

**Answer: A. Cortex**

The arrows point to the **round follicles lined up just under the capsule**, some with pale germinal centers. Follicles sit in the **outer cortex**, the **B-cell zone**.
**B (paracortex):** the diffuse, **follicle-free** tissue **below** the follicles; the T-cell zone with HEVs (high endothelial venules).
**C (medulla):** the deepest part, toward the hilum, with **dark cords** and **pale sinuses**.
**Exam twist:** Immunostain version: **CD20** (cluster of differentiation 20, a B-cell marker) turns the cortical follicles brown; **CD3** (a T-cell marker) turns the paracortex brown.

**34.** *2026 lymphoid lab deck*
What lymph node region is identified by the arrows?

Q7- What lymph node region is identified by the arrows?



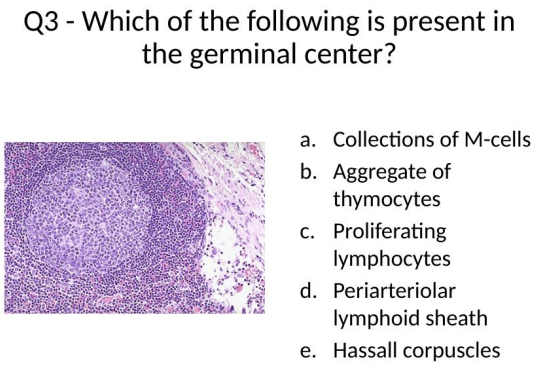
a. Cortex
b. Paracortex
c. Medulla

**A.** Cortex
**B.** Paracortex
**C.** Medulla

**Answer: B. Paracortex**

The arrows pass **beyond the follicles** and stop in the diffuse, **follicle-free** lymphoid tissue between the follicles and the medulla = **paracortex** (deep cortex). It is the **T-cell zone**, where HEVs (high endothelial venules) let blood lymphocytes in.
**A (cortex):** the arrows would stop on the **round follicles** right under the capsule (previous question).
**C (medulla):** deeper still, toward the hilum at the bottom of the picture, with dark branching cords and pale sinuses.
**Exam twist:** Clinical version: with no thymus (**DiGeorge syndrome**), the **paracortex** and the splenic PALS (periarteriolar lymphoid sheath) are depleted, while B-cell follicles stay normal.

**35.** *2026 lymphoid session*
Which of the following is present in the germinal center?

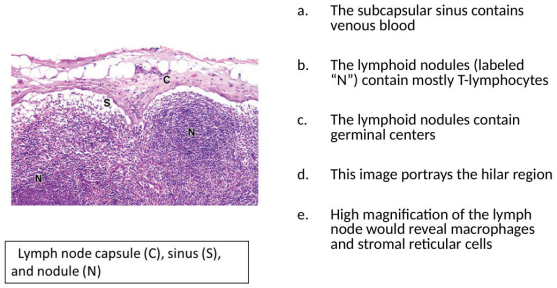



**A.** Collections of M-cells
**B.** Aggregate of thymocytes
**C.** Proliferating lymphocytes
**D.** Periarteriolar lymphoid sheath
**E.** Hassall corpuscles

**Answer: C. Proliferating lymphocytes**

The picture shows a follicle with a **pale center** capped by a **dark mantle** of small resting B cells. The pale **germinal center** is full of **large, activated, dividing B lymphocytes** with open (euchromatic) nuclei, resting on follicular dendritic cells. It forms only after antigen exposure (a secondary follicle).
**A (M cells):** epithelial cells over **Peyer's patches** that sample gut antigen.
**B (thymocytes):** developing T cells in the **thymus**, which has no germinal centers.
**D (PALS):** the PALS (periarteriolar lymphoid sheath) is the T-cell cuff around the central arteriole in **spleen** white pulp.
**E (Hassall corpuscles):** pink concentric whorls in the **thymic medulla**.
**Exam twist:** Expect **tingible body macrophages** (macrophages eating dying B cells in reactive germinal centers), or "primary follicle = no germinal center; secondary = germinal center + mantle".

**36.** *2026 lymphoid session*
Which of the following is true of the lymph node in this picture?



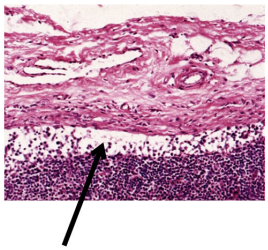

**A.** The subcapsular sinus contains venous blood
**B.** The lymphoid nodules (labeled "N") contain mostly T-lymphocytes
**C.** The lymphoid nodules contain germinal centers
**D.** This image portrays the hilar region
**E.** High magnification of the lymph node would reveal macrophages and stromal reticular cells

**Answer: E. High magnification of the lymph node would reveal macrophages and stromal reticular cells**

The picture shows the capsule (**C**), the **subcapsular sinus (S)** under it, and outer-cortex nodules (**N**). Node sinuses are crossed by a mesh of **reticular cells** and fibers and hold **macrophages** that filter the lymph; high power shows both.
**A (venous blood in the sinus):** lymph node sinuses carry **lymph**, not blood.
**B (nodules mostly T cells):** nodules are **B-cell** zones. T cells sit in the paracortex.
**C (nodules have germinal centers):** these nodules are evenly dark with **no pale center**, so they are **primary nodules** (the lab deck labels this same image that way).
**D (hilar region):** a convex capsule with a subcapsular sinus and nodules is the **cortex**. The hilum is the concave notch where the efferent lymphatic leaves.
**Exam twist:** Spleen contrast: the spleen has a capsule but **no subcapsular sinus** and no afferent lymphatics.

**37.** *2026 lymphoid lab deck*
What lymph node sinus is indicated by the arrow?

Q10- What lymph node sinus is indicated by the arrow?



a. Subcapsular sinus
b. Trabecular sinus
c. Medullary sinus

**A.** Subcapsular sinus
**B.** Trabecular sinus
**C.** Medullary sinus

**Answer: A. Subcapsular sinus**

The arrow points to the **pale, loose band right under the thick pink capsule**, above the dense purple cortex. That first space is the **subcapsular sinus**, where afferent lymph enters the node.
**B (trabecular sinus):** runs **inward alongside a trabecula** through the cortex, at right angles to the capsule.
**C (medullary sinus):** pale channels **between dark medullary cords**, deep in the node near the hilum.
**Exam twist:** Clinical version: cancer cells arrive in afferent lymph and lodge **first in the subcapsular sinus** of the sentinel node.

**38.** *2026 TA review*
Which sequence correctly traces the pathway of lymph flowing through a lymph node?

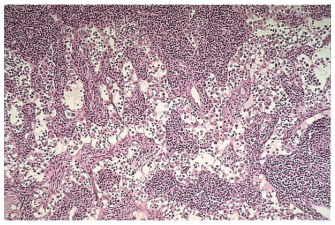
**A.** Efferent vessel -> Medullary sinus -> Paracortex -> Subcapsular sinus -> Afferent vessel
**B.** Subcapsular sinus -> Cortical (trabecular) sinuses -> Medullary sinuses -> Efferent lymphatic vessel
**C.** High endothelial venule -> Germinal center -> Medullary cord -> Trabecular vein
**D.** Medullary sinus -> Subcapsular sinus -> Cortical sinus -> Hilum

**Answer: B. Subcapsular sinus -> Cortical (trabecular) sinuses -> Medullary sinuses -> Efferent lymphatic vessel**

Lymph enters by **afferent vessels** on the **convex capsule**, fills the subcapsular sinus, runs down cortical (trabecular) sinuses beside the trabeculae, collects in medullary sinuses, and leaves by the **efferent vessel at the hilum**. Outside → in.
**A (efferent → … → afferent):** the correct path run **backwards**.
**C (HEV → germinal center → medullary cord → trabecular vein):** HEVs (high endothelial venules) carry **blood**, not lymph; this chain mixes blood vessels with tissue zones.
**D (medullary → subcapsular → cortical → hilum):** starts in the medulla. Lymph enters at the **capsule**.
**Exam twist:** "How do lymphocytes enter a node?" Two ways: **afferent lymph** and **HEVs from the blood**. They leave only by the **efferent lymphatic**.

**39.** *2026 lymphoid lab deck*
Which type of lymph node sinus is demonstrated in this picture?

Q11- Which type of lymph node sinus is demonstrated in this picture?



a. Subcapsular sinus
b. Trabecular sinus
c. Medullary sinus

**A.** Subcapsular sinus
**B.** Trabecular sinus
**C.** Medullary sinus

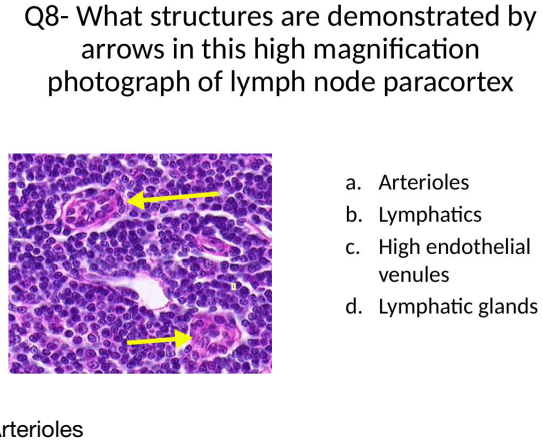
**Answer: C. Medullary sinus**

The field is full of **pale, branching channels** holding loose cells (macrophages, lymphocytes) that run **between dark, cell-packed cords**. Dark cords + pale sinuses in a net = **medulla**, so the pale channels are **medullary sinuses**.
**A (subcapsular sinus):** a **single thin pale band** right under the capsule.
**B (trabecular sinus):** one channel running **beside a pink trabecula** through the cortex.
**Exam twist:** "Where are plasma cells in a lymph node?" = the dark **medullary cords**. "**Sinus histiocytosis**" = medullary sinuses widened and packed with macrophages.

## Lymph node histology, HEVs, plasma cells

**40.** *2026 lymphoid lab deck*

What structures are demonstrated by arrows in this high magnification photograph of lymph node paracortex




**A.** Arterioles

**B.** Lymphatics

**C.** High endothelial venules

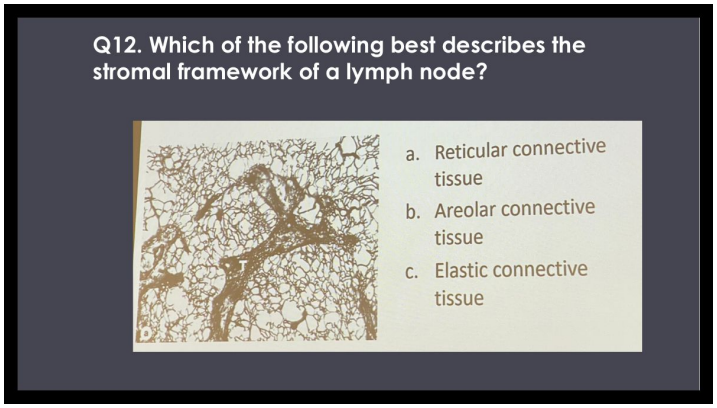
**D.** Lymphatic glands

**Answer: C. High endothelial venules**

In the paracortex, the yellow arrows mark small vessels lined by **plump, cuboidal ("high") endothelial cells** instead of flat ones = **HEVs (high endothelial venules)**. This is where **blood lymphocytes enter** the node, crossing the venule wall.
**A (arterioles):** have a ring of **smooth muscle** and flat endothelium.
**B (lymphatics):** very **thin walls**, flat endothelium, wide pale lumen.
**D (lymphatic glands):** not a real histologic structure ("lymph gland" is a lay term for a node).
**Exam twist:** "Where are HEVs?" = the **paracortex** (T-cell zone). They also occur in tonsils and Peyer's patches, but **not in the spleen**.

**41.** *2026 lymphoid lab deck*

Which of the following best describes the stromal framework of a lymph node?




**A.** Reticular connective tissue

**B.** Areolar connective tissue

**C.** Elastic connective tissue

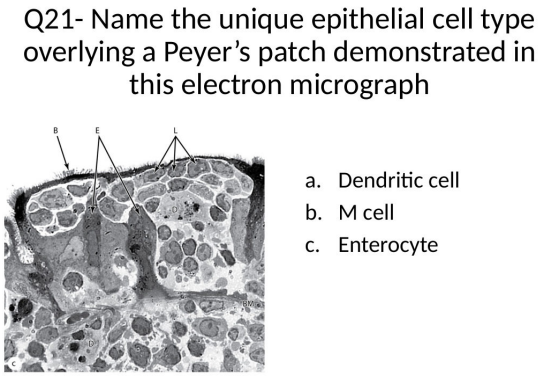
**Answer: A. Reticular connective tissue**

This is a **silver stain**, which turns **reticular fibers** (type III collagen) **black**. The fine black **three-dimensional mesh** is **reticular connective tissue**, the stroma of lymph nodes (also spleen and marrow). **T** marks a thicker trabecula.
**B (areolar connective tissue):** loose tissue under epithelia with mixed collagen and elastic fibers and many cell types; not a fine, even mesh.
**C (elastic connective tissue):** found in large arteries and some ligaments; it needs an elastic stain and forms sheets or thick fibers.
**Exam twist:** Exception question: the **thymus** has **no reticular fibers**. Its framework is made of **epithelial** reticular cells (TECs, thymic epithelial cells).

## MALT: tonsils, Peyer's patches, appendix (incl. IDs)

**42.** *2026 lymphoid lab deck*

Name the unique epithelial cell type overlying a Peyer's patch demonstrated in this electron micrograph




**A.** Dendritic cell

**B.** M cell

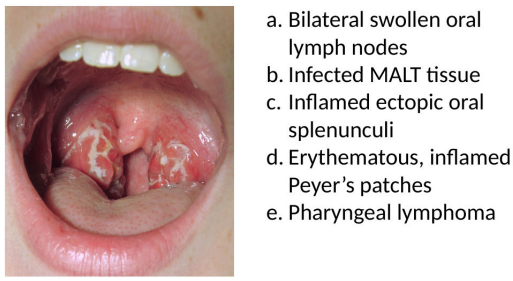
**C.** Enterocyte

**Answer: B. M cell**

On the dome over a Peyer's patch, the epithelial cells that **cap pale pockets full of lymphocytes (L)** and **dendritic cells (D)** are **M cells**. They have **no brush border**, only short folds, and thin cytoplasm wrapped around a **basal pocket**. They take up gut antigen and hand it to the immune cells in the pocket.
**A (dendritic cell):** the D cells sit **inside the pocket and below** the basement membrane (BM); they are immune cells, not epithelial cells.
**C (enterocyte):** the tall dark absorptive cells (**E**) between pockets, topped by a **brush border** of microvilli (**B**).
**Exam twist:** "Which cell carries antigen across the gut epithelium?" = **M cell**. Nearby plasma cells then make **IgA** (immunoglobulin A).

**43.** *2026 lymphoid session*

A 10-year-old boy presented to his doctor with fever, sore throat, and tender, swollen cervical lymph nodes. The doctor examined the oral cavity and prescribed antibiotics. Which of the following best applies to the abnormalities visible on examination of the oral cavity?




**A.** Bilateral swollen oral lymph nodes

**B.** Infected MALT tissue

**C.** Inflamed ectopic oral splenunculi

**D.** Erythematous, inflamed Peyer's patches

**E.** Pharyngeal lymphoma

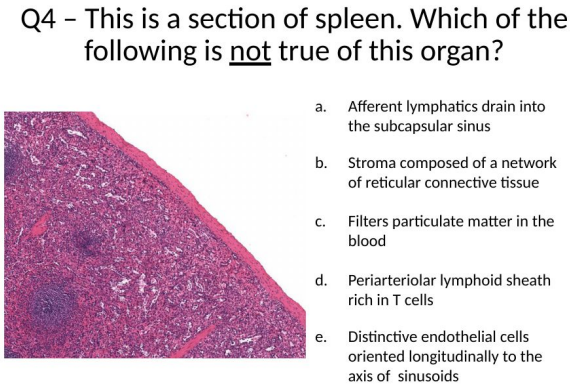
**Answer: B. Infected MALT tissue**

Fever, sore throat, tender neck nodes and **red, swollen palatine tonsils with white exudate** = **tonsillitis**. Palatine tonsils are **MALT**: partly encapsulated nodules under stratified squamous epithelium with deep **crypts** that trap bacteria and debris.
**A (oral lymph nodes):** no lymph nodes sit in the oral mucosa. The swollen nodes are **cervical** (in the neck) and drain the tonsils.
**C (ectopic splenunculi):** splenunculi are small extra spleens near the spleen, not in the mouth.
**D (Peyer's patches):** MALT of the **ileum**, not the throat.
**E (pharyngeal lymphoma):** sudden fever and exudate in a child, treated with antibiotics, point to **infection**, not a tumor.
**Exam twist:** Histology version: **palatine** tonsil = stratified squamous epithelium + deep crypts; **pharyngeal** tonsil (adenoid) = respiratory epithelium, no crypts.

## Spleen structure: white and red pulp (incl. IDs)

**44.** *2025 faculty lymphoid session*

This is a section of spleen. Which of the following is not true of this organ?



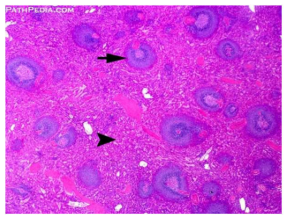

**A.** Afferent lymphatics drain into the subcapsular sinus
**B.** Stroma composed of a network of reticular connective tissue
**C.** Filters particulate matter in the blood
**D.** Periarteriolar lymphoid sheath rich in T cells
**E.** Distinctive endothelial cells oriented longitudinally to the axis of sinusoids

**Answer: A. Afferent lymphatics drain into the subcapsular sinus**

The spleen filters **blood**, not lymph. It has **no afferent lymphatics** and **no subcapsular sinus**: in the picture the thick capsule sits right on **red pulp**. Afferent lymphatics emptying into a subcapsular sinus describes a **lymph node**.
**B (reticular stroma):** true. Reticular cells and fibers form the framework.
**C (filters particles from blood):** true. It removes old RBCs (red blood cells), bacteria and debris.
**D (PALS rich in T cells):** true. The PALS (periarteriolar lymphoid sheath) is the T-cell cuff around the central arteriole.
**E (lengthwise endothelial cells):** true. **Stave cells** line the sinusoids like barrel staves.
**Exam twist:** Organ-ID version: lymph node = subcapsular sinus + cortex, paracortex, medulla; spleen = no subcapsular sinus, white pulp nodules with an off-center arteriole scattered in red pulp.

**45.** *2026 lymphoid lab deck*

The splenic structure demonstrated by the arrow is…



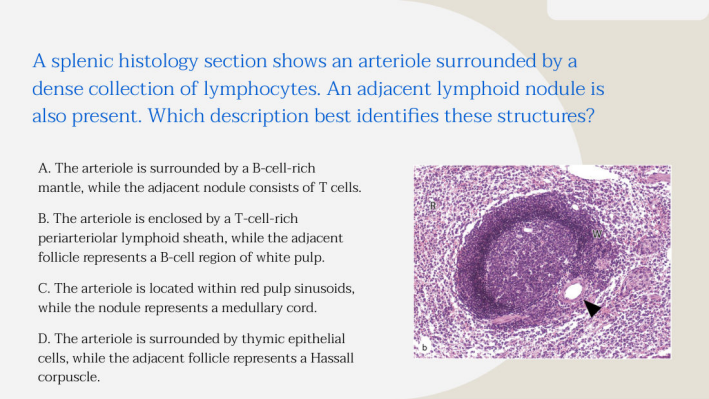

**A.** White pulp
**B.** Red pulp
**C.** White matter
**D.** Gray matter
**E.** Trabeculum

**Answer: A. White pulp**

At low power the arrow sits on a **round, dark purple nodule**, many with a pale center and a small artery = **white pulp** (lymphoid tissue: PALS, the periarteriolar lymphoid sheath, plus a B-cell follicle). The pink-red tissue around the nodules is red pulp.
**B (red pulp):** the **pink tissue between** the nodules (where the arrowhead points), full of blood in cords and sinusoids.
**C (white matter):** a brain and spinal cord term (myelinated axons), not spleen.
**D (gray matter):** also a nervous system term (neuron cell bodies).
**E (trabeculum):** the **smooth, pale pink** connective tissue bands carrying vessels, with no lymphocytes.
**Exam twist:** "Which zone removes old RBCs (red blood cells)?" = **red pulp**, which enlarges in splenomegaly from hemolysis; "which zone mounts immune responses to blood antigens?" = white pulp.

**46.** *2026 TA review*

A splenic histology section shows an arteriole surrounded by a dense collection of lymphocytes. An adjacent lymphoid nodule is also present. Which description best identifies these structures?




**A.** The arteriole is surrounded by a B-cell-rich mantle, while the adjacent nodule consists of T cells.

**B.** The arteriole is enclosed by a T-cell-rich periarteriolar lymphoid sheath, while the adjacent follicle represents a B-cell region of white pulp.

**C.** The arteriole is located within red pulp sinusoids, while the nodule represents a medullary cord.

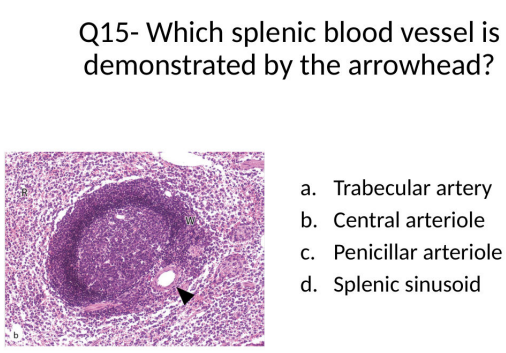
**D.** The arteriole is surrounded by thymic epithelial cells, while the adjacent follicle represents a Hassall corpuscle.

**Answer: B. The arteriole is enclosed by a T-cell-rich periarteriolar lymphoid sheath, while the adjacent follicle represents a B-cell region of white pulp.**

The arrowhead marks the **central arteriole** at the edge of white pulp (**W**). The lymphocytes hugging it form the **PALS** (periarteriolar lymphoid sheath) = **T cells**. The large round nodule beside it is the **B-cell follicle**. **R** = red pulp.
**A (B-cell mantle around arteriole, T-cell nodule):** swaps the two. **T** cells hug the arteriole; **B** cells fill the nodule.
**C (red pulp sinusoids; medullary cord):** the arteriole is in white pulp, and medullary cords belong to a **lymph node**.
**D (thymic epithelial cells; Hassall corpuscle):** these exist only in the **thymus**.
**Exam twist:** Node-to-spleen matching: **PALS ≈ paracortex** (T cells), **splenic follicle ≈ cortical follicle** (B cells). With no thymus (DiGeorge syndrome) the PALS is thin.

**47.** *2026 lymphoid lab deck*

Which splenic blood vessel is demonstrated by the arrowhead?




**A.** Trabecular artery

**B.** Central arteriole

**C.** Penicillar arteriole

**D.** Splenic sinusoid

**Answer: B. Central arteriole**

The arrowhead marks a small, round, **muscular arteriole inside the white pulp (W)**, at the edge of the lymphoid nodule. An arteriole wrapped in lymphocytes in white pulp is the **central arteriole**; it is usually **off-center**, despite the name.
**A (trabecular artery):** larger, and sits inside a **pink connective tissue trabecula**, not in lymphoid tissue.
**C (penicillar arteriole):** small straight branches out in the **red pulp (R)**, after the central arteriole leaves the white pulp.
**D (splenic sinusoid):** a wide, thin-walled channel lined by stave cells in the red pulp, with no muscle wall.
**Exam twist:** Blood path order: **trabecular artery** → **central arteriole** → **penicillar arterioles** → **sheathed capillaries** → cords (open) or sinusoids (closed) → **trabecular vein**.

## Splenic blood flow and RBC removal

**48.** *2026 TA review*

In the open theory of splenic circulation, where is blood discharged from the penicillar arterioles?

**A.** Directly into the lumen of the splenic vein

**B.** Into the central arteriole of the white pulp

**C.** Into the subcapsular sinus of the spleen

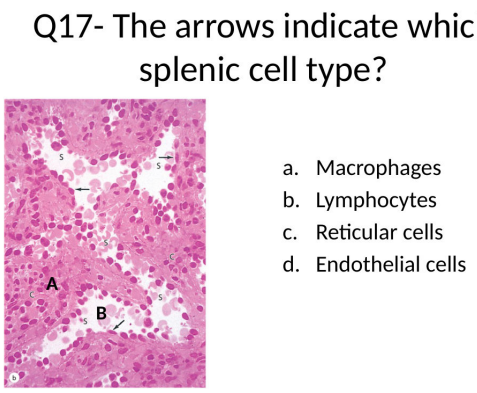
**D.** Directly into the meshwork/stroma of the splenic cords (cords of Billroth)

**Answer: D. Directly into the meshwork/stroma of the splenic cords (cords of Billroth)**

In **open** circulation, penicillar arterioles and sheathed capillaries **end open** in the red pulp and dump blood into the **splenic cords**. Cells must then squeeze **between stave cells** back into the sinusoids. Old, rigid RBCs (red blood cells) can't pass and are eaten by cord macrophages.
**A (splenic vein lumen):** the splenic vein comes **last**, after sinusoids → red pulp veins → trabecular veins.
**B (central arteriole):** comes **before** the penicillar arterioles, in the white pulp.
**C (subcapsular sinus):** the spleen has **no subcapsular sinus**; that is a lymph node feature.
**Exam twist:** Flip: in **closed** circulation the capillaries connect **straight into the sinusoids**. Sickle cells and spherocytes get trapped in the cords through open circulation.

**49.** *2026 lymphoid lab deck*

The arrows indicate which splenic cell type?




**A.** Macrophages

**B.** Lymphocytes

**C.** Reticular cells

**D.** Endothelial cells

**Answer: D. Endothelial cells**

The arrows mark **elongated nuclei that line the open sinusoid lumens (S)** and **bulge into them**. These are the endothelial cells of splenic sinusoids, called **stave cells**: long cells running lengthwise like barrel staves, with slits between them.
**A (macrophages):** big cells with pale cytoplasm, often holding eaten RBCs (red blood cells), sitting **in the cords (C)**, not lining lumens.
**B (lymphocytes): small, round, dark** nuclei scattered in the cords and lumens.
**C (reticular cells):** stromal cells that make reticular fibers **inside the cords**; they don't line the lumen.
**Exam twist:** Silver stain version: black **ring fibers** wrap the sinusoids like barrel hoops. In open circulation RBCs must squeeze **between stave cells**, and rigid ones get trapped.

## Clinical correlates: metastasis, splenomegaly, integration

**50.** *2026 lymphoid session*

A 47-year-old woman detected a cancerous lump in the upper outer quadrant of her breast. Her physician recommended surgical removal of the mass with a sentinal lymph node biopsy of the first axillary lymph node that the cancer was most likely to spread to from the primary tumor. If pathologic examination failed to reveal tumor in the sentinal lymph node, is the likelihood of finding positive lymph nodes in the remaining axillary lymph nodes increased or decreased?.

**A.** Increased likelihood of positive axillary lymph nodes

**B.** Decreased likelihood of positive axillary lymph nodes

**Answer: B. Decreased likelihood of positive axillary lymph nodes**

The **sentinel node** is the **first** node to receive lymph from the tumor, so cancer cells traveling in afferent lymph lodge there **first**. If the sentinel node is **negative**, spread to the other axillary nodes is **less likely**, and a full axillary dissection can often be skipped.
**A (increased likelihood):** backwards. A positive sentinel node raises the chance of other positive nodes; a negative one lowers it.
**Exam twist:** Asked as "where in the node do metastatic cells appear first?" = **subcapsular sinus**; or "how is the sentinel node found?" = injected **blue dye or radioactive tracer** follows the lymph.